Acid Drip - Quick Reference (V5, this build)

Bassline synth, DRIFT synth and drum machine. Wiring: Sync IN/OUT=GP8 - TFT RST=3V3 - Acid=GP15 - Drums+DRIFT=GP2

Acid Pad Layout

Top row steps 1-8; bottom row 9-16 (labels KEY/RIFF/SOUND/WALK/FX/TEMPO/PLEN/PAT> show in FUNC mode).

Pad	Function	Pad	Function
9 KEY	Root note	13 FX	FX assign sub-mode
10 RIFF	Preset riff	14 TEMPO	Preset BPM / tap tempo
11 SOUND	Synth voice	15 PLEN	Pattern length 1-16
12 WALK	Note walk	16 PAT>	Playback order

Step Editing

Action	Result	Action	Result
Short press pad	Toggle step on/off	Long press x2	Toggle glide
Long press x1	Toggle accent	Hold pad + CUT pot	Set step note

Core Chords

Chord	Action
1+2	Play / Stop
1+2 hold 0.5s	Reset current engine (acid/DRIFT). Drums: hold 1s in drum mode
7+8	Enter / exit FUNC mode
9+10	Switch to drums, or back to acid/DRIFT
9+10+11 hold 0.5s	Acid <-> DRIFT. No-op from drums
11+12+13+14 hold 1s	Acid Walks easter egg
12+13	Enter / exit chain-build mode
15+16 hold 1s	Toggle MIX EDIT mode

Save / Load (hold 7+8)

Tap 3/4/5/6 = load slot 1-4. Hold 3/4/5/6 1s = save. Hold 3+4+5+6 1s = clear all 4 slots (per engine). Each slot saves acid, DRIFT and drums together plus play/stop state, from any screen.

FUNC Mode (hold 7+8)

Pad	Function	Options
9	KEY	C D Eb F G Ab Bb B
10	RIFF	DFLT / SQNCE / FUNK / MINI / JUMP / RAVE / SYNC / DARK
11	SOUND	SAW / SQR / SINE / PWM / CSAW / CSQR / CSIN / SUBSQ
12	WALK	OFF / 4TH / OCTWAVE / 5TH / BOUNCE / MIN3RD / VAMP3 / RANDOM
13	FX	Enter FX assign sub-mode
14	TEMPO	Preset BPM or tap tempo
15	PLEN	Pattern length 1/2/3/4/6/8/12/16
16	PAT>	FWD / CW / ALT / REV / SKIP2 / SKIP3 / PING / RND

FX Assign (FUNC, pad 13): top-row pad picks effect (None/Oct Up/Retrigger/Stutter/Maj Step/Min Step/Dom7 Step/Dim Step), then a step pad assigns it. Tap effect pad to deselect. Long-hold pad 1 clears all.

Pattern Chaining

Tap 12+13 = chain-build mode; tap 3/4/5/6 to append that slot (repeat = repeat); tap 12+13 again to exit and play. Each position plays one full loop of its slot length. Acid only, not drum mode. Not saved to EEPROM.

MIX EDIT

Hold 15+16 for 1s to trim each engine's level independently. Hold again to exit; levels auto-save to EEPROM.

Pot	**Engine**
CUT	Acid
RES	DRIFT
DCY	Drums

DRIFT Synth - switching

Second synth voice, sums into the drums output (GP2).

Currently on	**9+10**	**9+10+11 (hold 0.5s)**
Acid	Go to drums	Go to DRIFT
DRIFT	Go to drums	Back to acid
Drums	Back to acid/DRIFT	No-op

DRIFT reachable only from acid screen. All three engines can play at once; switching only changes which one the pads/pots control.

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DRIFT FUNC Mode (7+8)

Pad	Function	Options
9	SND1	SUBSIN / DTSQR / FMBEL / RINGM / CLICK / FOLDR / PWM / UNISN
10	SND2	SAW / SUBDIV / NOIZ / VOWEL / ARPG / GONG / CRUSH / GATE
11	FOLW	F+8 / F-8 / F+5 / F+3 / FOLW / ARP+ / AR+- / WALK
12	ECHO	Delay: OFF / 8D / 1/4 / 4D / 1/2 / 8FB / 4FB / 2FB
13	VERB	Reverb: OFF / SML / MED / BIG / HALL / MEDW / BIGW / WASH
14	PATT	8 pattern slots S1-S8 (tap load, hold save)
15	EVOL	OFF / OCTL / ADD1 / DEEP / STUT / HVY / WILD / MAX
16	REC	Pot motion recorder (tap start/stop, hold clear)

Beat Machine

Switch with pads 9+10. Sequencer state preserved across engines.

Pad	Instrument	Pad	Instrument
1	Kick	5	Tom
2	Hi-hat (closed)	6	Bass 2
3	Snare	7	Clap
4	Rim	8	Open hat

Short-tap pad 1-16 loads that pattern. Pads 1-5 double as edit pads (tap = select, long hold = edit). 6-8 select-only.

Per-Drum Editing (hold pad 1-5)

Pot	Control
CUT	Pitch
RES	Decay length
DCY	Volume

Custom Pattern (Pad 16)

Action	Result
Quick tap	Enter custom edit mode
Tap while editing	Toggle step 16 for current voice
Turn any pot	Set pitch/length/volume for current voice
Hold 0.5s	Cycle to next drum voice
Hold 1s	Exit and save
Hold 1.5s	Wipe pattern and restart

Beat Machine FUNC (7+8)

Label	Function
PTCH	Global pitch, stacks on per-drum
DRIV	Drive / saturation, bit crush at top
FILT	Bipolar DJ filter (low/high sweep)

Label	Function
CHNC	Chance - thin or add ghost hits
HMNZ	Humanize - velocity & timing jitter
SWNG	Swing on odd 16th steps
ACNT	Accent depth, on/off-beat emphasis
FILL	Procedural fill engine

Fill Engine

Pads 1-4 frequency: OFF / 1 BAR / 2 BAR / 4 BAR. Pads 5-8 type: HATS / CLAP / SNARE / KIT. Pots control the fill voice's pitch/decay/volume while FILL selected. Density accelerates into the landing.

16 Preset Patterns

1 BASIC 2 HOUSE 3 TECHNO 4 ACID TRACK 5 HI STATE 6 FUNKY 7 SWING 8 BREAKBEAT 9 MINIMAL 10 LATIN 11 PACIFIC ST 12 BLUE MONDAY 13 DRUM N BASS 14 VOODOO RAY 15 BUILD DOWN 16 CUSTOM

Sync In / Out - this build (GP8)

Hold pad 14 at power-on for the SYNC MODE menu: pad 1 = Sync In (external clock on GPIO8 drives tempo), pad 2 = Sync Out (GPIO8 sends 24 PPQN DIN sync to other gear). Confirmation shows on screen. Sync Out pulses only while the acid transport runs and the info bar "S" indicator shows direction (cyan = In, magenta = Out).

	This build (GP8)
Sync pin	IN and OUT both on GPIO8 (same jack), direction chosen at boot
Drums / DRIFT	Stay live in BOTH sync directions on GP2 - syncing never silences them
TFT RST	Moved to 3V3 to free GPIO8 (GP8 was the original RST pin)

Acid Drip V5. Hardware and firmware by Marcus (lonesoulsurfer), github.com/lonesoulsurfer. Regenerated quick ref with GP8 sync corrections.